Chap.14 Frequency Selective Circuits [Filters]

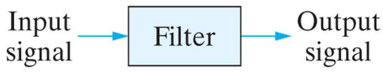
※ Chap. 12-13 Laplace Transform

$$X(s) = \int_0^{\infty} x(t)\, e^{-st}\, dt$$

where $s = \sigma + j\omega$ (complex number)

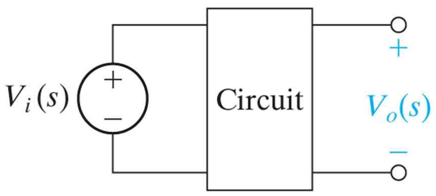
※ $s = j\omega$ in this part

- Frequency selective circuits are also called FILTERs.

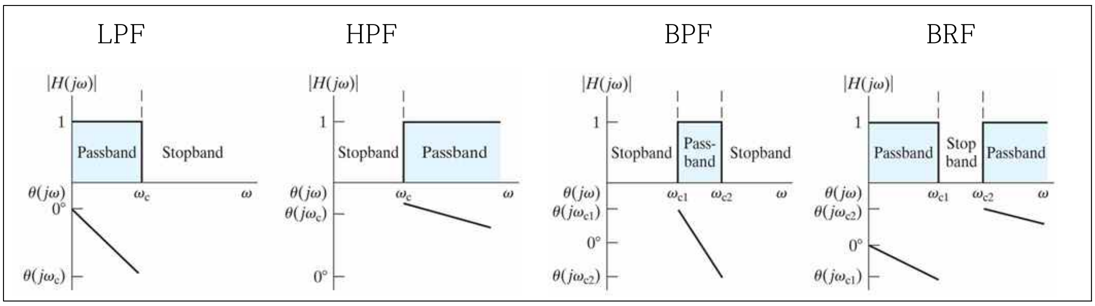


§14.1 Some preliminaries

- A circuit with voltage input and voltage output

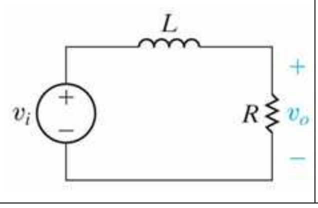
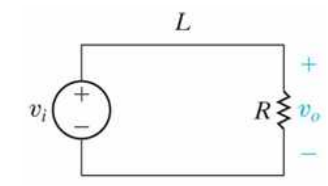
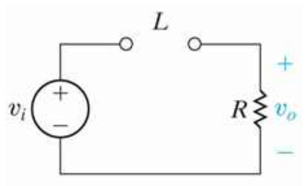


- Frequency response
- passband, stopband, transition band
- cutoff frequency, low-pass filter, high-pass filter,
  bandpass filter, bandreject (or bandstop) filter
- Ideal Filters

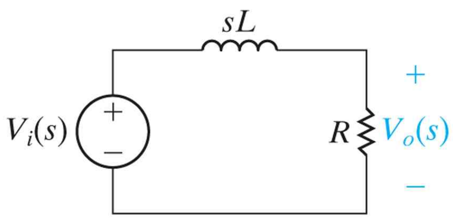


- passive filter, active filter

§14.2 Low-Pass Filters (LPF)

a series RL LPF	equivalent circuit ($\omega = 0$)	equivalent circuit ($\omega = \infty$)
		

- The series RL circuit – Quantitative Analysis



current $\quad I(j\omega) = \dfrac{V_i(j\omega)}{j\omega L + R}$

output voltage

$$V_0(j\omega) = I(j\omega) \times R = \frac{V_i(j\omega)}{j\omega L + R} \times R$$

the frequency response

$$H(j\omega) = \frac{V_0(j\omega)}{V_i(j\omega)} = \frac{R}{j\omega L + R} = \frac{R/L}{j\omega + R/L} \qquad \text{Eq.(\#)}$$

the magnitude plot and the angle plot of the frequency response

$$|H(j\omega)| = \frac{R/L}{\sqrt{\omega^2 + (R/L)^2}} \quad , \quad \theta(j\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right)$$

※ Defining the **Cutoff Frequency** (ω_c or f_c)

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}} H_{\max}$$

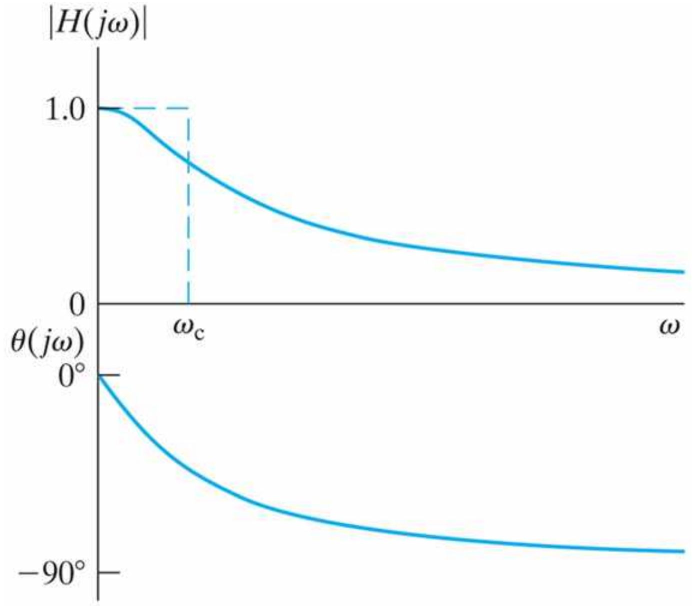
the cutoff frequency of the series RL circuit

$$H_{\max} = |H(j0)| = 1$$

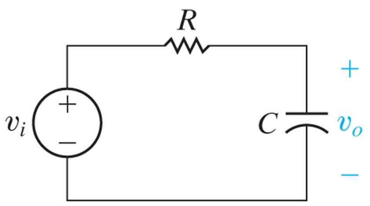
$$|H(j\omega_c)| = \frac{R/L}{\sqrt{\omega_c^2 + (R/L)^2}} = \frac{1}{\sqrt{2}} |1| \qquad \therefore \ \omega_c = \frac{R}{L}$$

※ time constant : $\tau = \dfrac{L}{R}$

- Frequency response of an RL LPF



- A series RC circuit



$$I(j\omega) = \frac{V_i(j\omega)}{R + \dfrac{1}{j\omega C}}$$

$$V_0(j\omega) = I(j\omega) \times \frac{1}{j\omega C} = \frac{V_i(j\omega)}{R + \dfrac{1}{j\omega C}} \times \frac{1}{j\omega C}$$

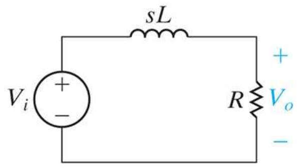
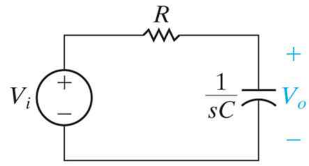
the <u>frequency response</u>

$$H(j\omega) = \frac{\dfrac{1}{j\omega C}}{R + \dfrac{1}{j\omega C}} = \frac{1}{j\omega RC + 1} = \frac{\dfrac{1}{RC}}{j\omega + \dfrac{1}{RC}} \qquad \text{Eq.(##)}$$

(time constant : $\tau = RC$)

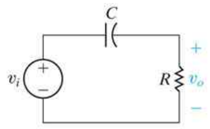
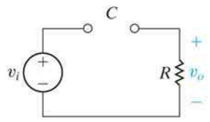
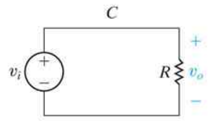
cutoff frequency : $\omega_c = \dfrac{1}{RC}$

※ Comparison [Low Pass Filters]

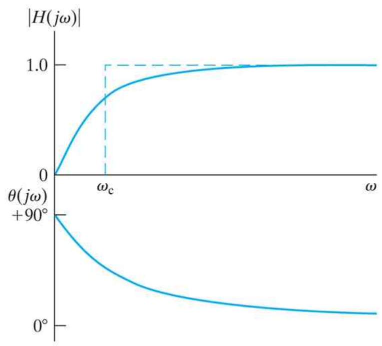
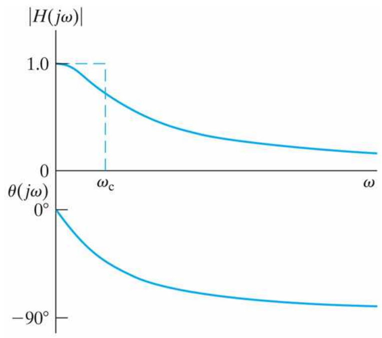
equation	Eq.(#)	Eq.(##)
RL or RC	RL circuit	RC circuit
circuit		
Frequency Response	$H(j\omega) = \dfrac{R/L}{j\omega + R/L}$	$H(j\omega) = \dfrac{\frac{1}{RC}}{j\omega + \frac{1}{RC}}$
Time Constant	$\tau = \dfrac{L}{R}$ [sec]	$\tau = RC$ [sec]
Cutoff Frequency	$\omega_c = \dfrac{R}{L}$ [rad/sec] (or $f_c = \dfrac{1}{2\pi}\left(\dfrac{R}{L}\right)$ [Hz])	$\omega_c = \dfrac{1}{RC}$ [rad/sec] (or $f_c = \dfrac{1}{2\pi}\left(\dfrac{1}{RC}\right)$ [Hz])
Remark	$H(j\omega) = \dfrac{constant}{j\omega + constant} = \dfrac{\omega_c}{j\omega + \omega_c}$ 1. General form of first-order LPF 2. one memory circuits 3. first-order differential equation in time domain 4. frequency response in frequency domain	

§14.3 High-Pass Filters (HPF)

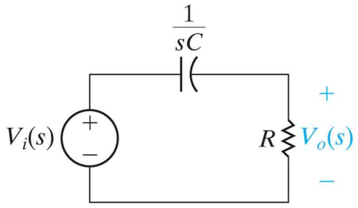
- A series RC circuit

a series RC HPF	equivalent circuit ($\omega = 0$)	equivalent circuit ($\omega = \infty$)
		

- Frequency response of the first-order RC HPF

frequency response of the 1st-order HPF	frequency response of the 1st-order LPF
	

- the first-order RC HPF



$$I(j\omega) = \frac{V_i(j\omega)}{\frac{1}{j\omega C} + R}$$

$$V_0(j\omega) = I(j\omega) \times R = \frac{V_i(j\omega)}{\frac{1}{j\omega C} + R} \times R = \frac{R}{\frac{1}{j\omega C} + R} \times V_i(j\omega)$$

$$= \frac{j\omega RC}{1 + j\omega RC} \times V_i(j\omega) = \frac{j\omega}{j\omega + \frac{1}{RC}} \times V_i(j\omega)$$

- the frequency response

$$H(j\omega) = \frac{V_0(j\omega)}{V_i(j\omega)} = \frac{j\omega}{j\omega + 1/RC} \qquad \text{Eq.}(*)$$

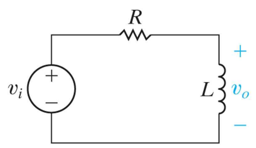
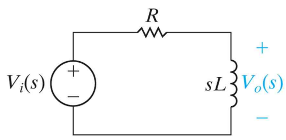
the magnitude and the angle of the frequency response

$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + (1/RC)^2}} \quad , \quad \theta(j\omega) = 90^\circ - \tan^{-1}\omega RC$$

the cutoff frequency : $\omega_c = \dfrac{1}{RC}$ (※ time constant : $\tau = RC$)

※ Same cutoff frequency (or time constant) with LPF

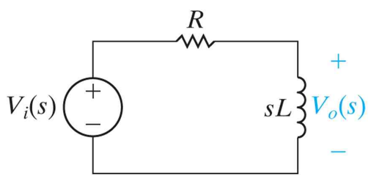
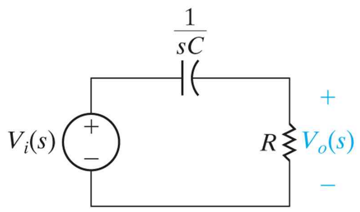
- (example) A series RL HPF

<table>
<tr><th>time-domain</th><th>frequency-domain</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

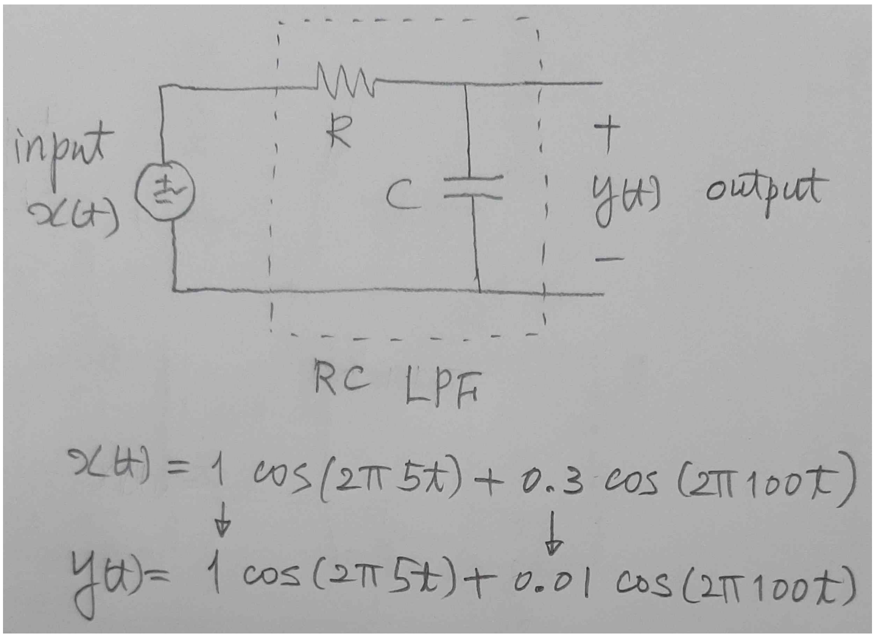
frequency response : $H(j\omega) = \dfrac{j\omega}{j\omega + R/L}$ Eq.(**)

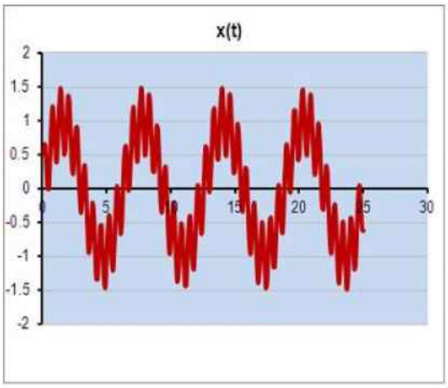
cutoff frequency : $\omega_c = \dfrac{R}{L}$

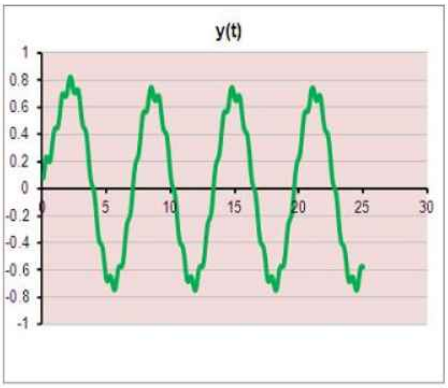
※ Comparison [High Pass Filters]

equation	Eq.(**)	Eq.(*)
RL or RC	RL circuit	RC circuit
circuit		
Frequency Response	$H(j\omega) = \dfrac{j\omega}{j\omega + R/L}$	$H(j\omega) = \dfrac{j\omega}{j\omega + 1/RC}$
Time Constant	$\tau = \dfrac{L}{R}$ [sec]	$\tau = RC$ [sec]
Cutoff Frequency	$\omega_c = \dfrac{R}{L}$ [rad/sec] (or $f_c = \dfrac{1}{2\pi}\left(\dfrac{R}{L}\right)$ [Hz])	$\omega_c = \dfrac{1}{RC}$ [rad/sec] (or $f_c = \dfrac{1}{2\pi}\left(\dfrac{1}{RC}\right)$ [Hz])
Remark	$H(j\omega) = \dfrac{j\omega}{j\omega + constant} = \dfrac{j\omega}{j\omega + \omega_c}$ 1. General form of first-order HPF 2. one memory circuits 3. first-order differential equation in time domain 4. frequency response in frequency domain	

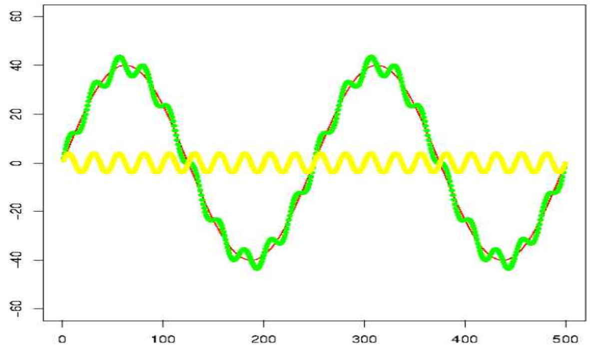
(Example) LPF (R-C Low Pass Filter)



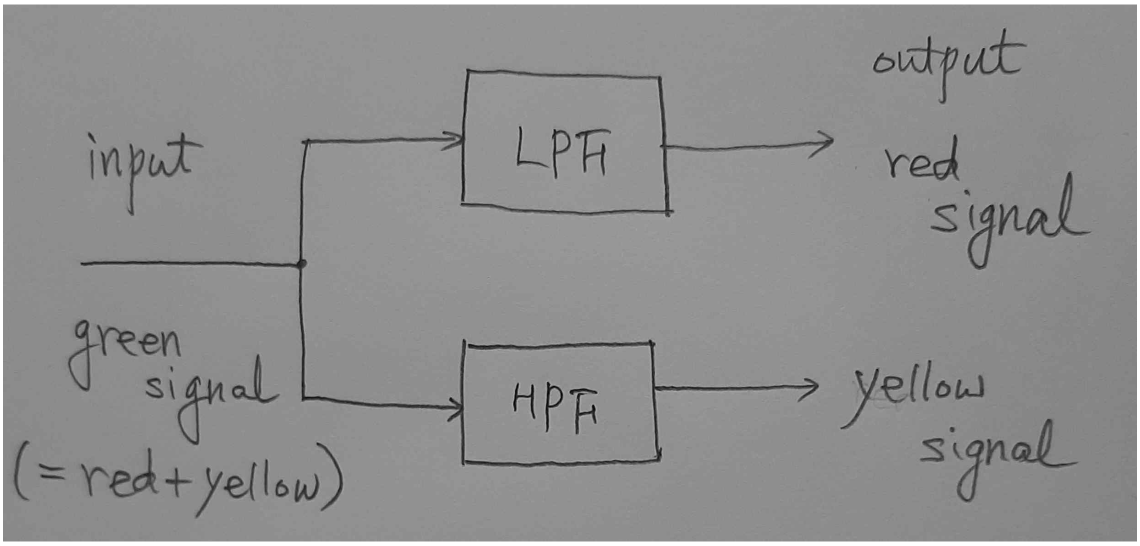




(Example) LPF, HPF



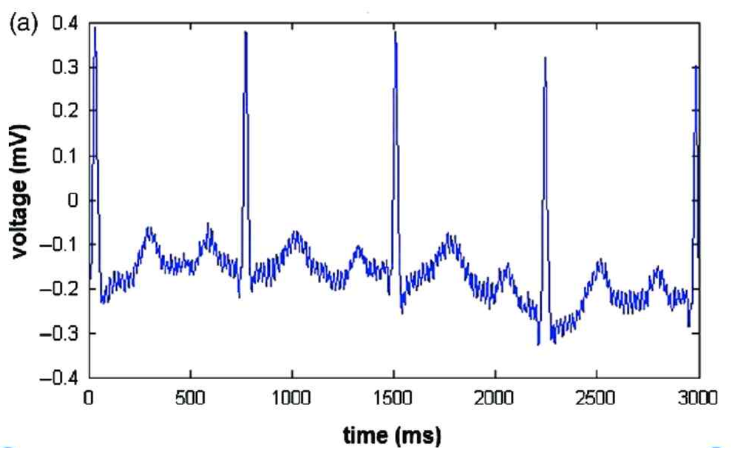
green signal = red signal (low frequency signal)
+ yellow signal (high frequency signal)



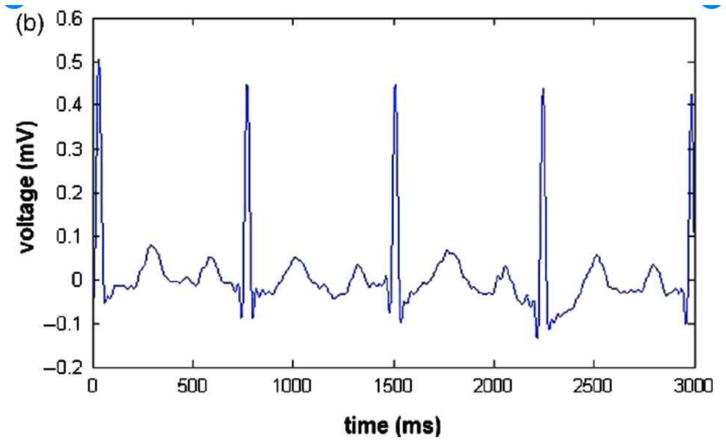
(Example of LPF) ECG (ElectroCardioGraph)

period T = 0.75 sec / heart rate = 80/minute

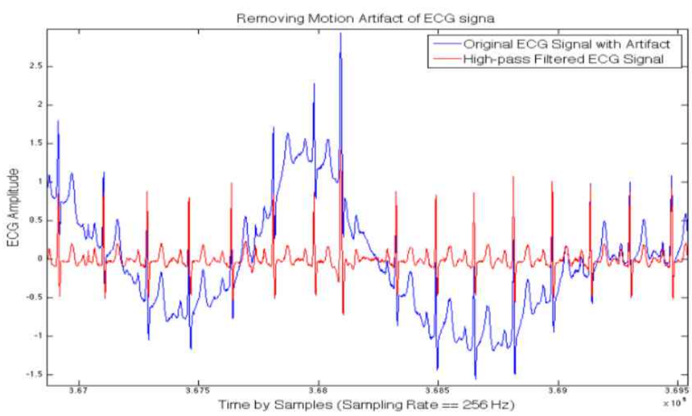
(a) measured signal



(b) filtered signal

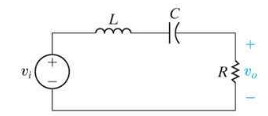
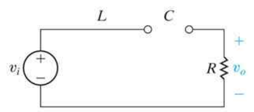
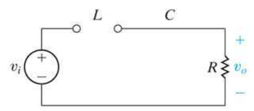


(Example of HPF) ECG (ElectroCardioGraph)

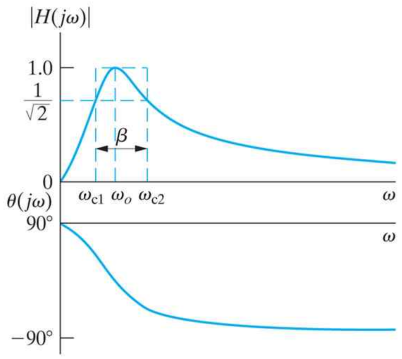
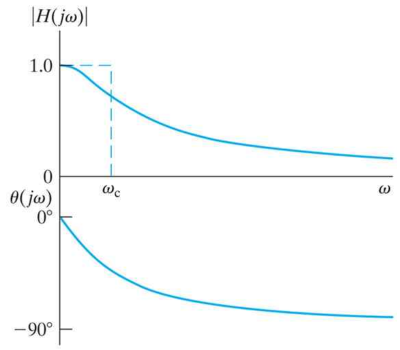


§14.4 Bandpass Filters (BPF)

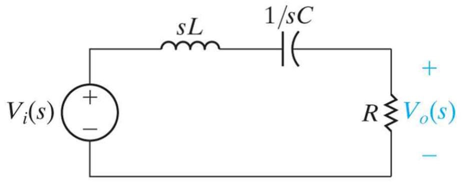
- A series RLC bandpass filter

a series RLC BPF	equivalent circuit ($\omega = 0$)	equivalent circuit ($\omega = \infty$)
		

- Frequency response of the second-order RLC BPF

frequency response of the 2nd-order BPF	frequency response of the 1st-order LPF
	

The series RLC Circuit



- frequency response

$$H(j\omega) = \frac{V_0(j\omega)}{V_i(j\omega)} = \frac{R}{R + j\omega L + \dfrac{1}{j\omega C}} = \frac{R}{R + j(\omega L - \dfrac{1}{\omega C})}$$

$$= \frac{(R/L)(j\omega)}{(j\omega)^2 + (R/L)(j\omega) + (1/LC)}$$

(magnitude) $$|H(j\omega)| = \frac{\omega(R/L)}{\sqrt{[(1/LC) - \omega^2]^2 + [\omega(R/L)]^2}}$$

(phase) $$\theta(j\omega) = 90^\circ - \tan^{-1}\left[\frac{\omega(R/L)}{(1/LC) - \omega^2}\right]$$

- center frequency : ω_0 [the frequency of max $|H(j\omega)|$]

  When $(1/LC)-\omega^2=0$, $|H(j\omega)|$ is maximum.

  $$\omega_0=\sqrt{\frac{1}{LC}} \qquad \text{(i.e., } f_0=\frac{1}{2\pi\sqrt{LC}} \text{ in physics class)}$$

  (At the center frequency, $j\omega_0 L+\dfrac{1}{j\omega_0 C}=0$.

  That is, at $\omega=\omega_0$, $Z=R+j\omega_0 L+\dfrac{1}{j\omega_0 C}=R$.

  So, when $\omega=\omega_0$, the RLC circuit will a resistor circuit.)

  ※ Resonant circuit in physics class

- cutoff frequencies : ω_{c1} and ω_{c2}

  $H_{\max}=|H(j\omega_0)|=1$

  frequencies ω_{c1} and ω_{c2} for $|H(j\omega_{c1})|=|H(j\omega_{c2})|=\dfrac{1}{\sqrt{2}}H_{\max}$

  $$\omega_{c1}=-\frac{R}{2L}+\sqrt{(\frac{R}{2L})^2+(\frac{1}{LC})}$$

  $$\omega_{c2}=\frac{R}{2L}+\sqrt{(\frac{R}{2L})^2+(\frac{1}{LC})}$$

- center frequency and cutoff frequencies

  $$\omega_0=\sqrt{\omega_{c1}\cdot\ \omega_{c2}}=\sqrt{\frac{1}{LC}}$$

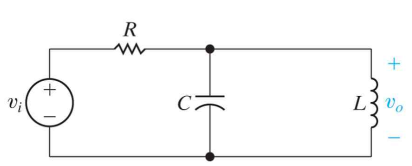
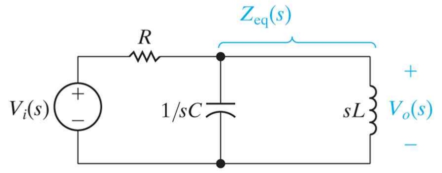
- bandwidth : β

  $$\beta=\omega_{c2}-\omega_{c1}=\frac{R}{L}$$

- Quality Factor : Q

  $$Q=\frac{\omega_0}{\beta}=\sqrt{\frac{L}{CR^2}}$$

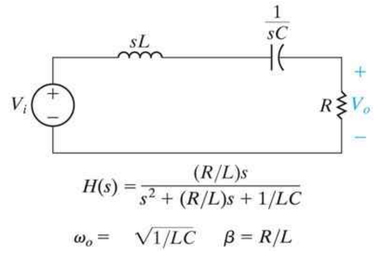
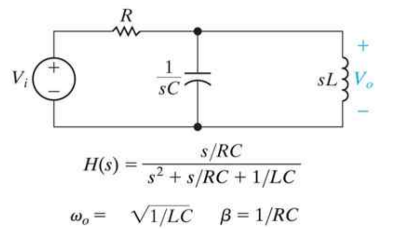
- a parallel RLC BPF

a parallel RLC BPF	the frequency-domain equivalent
	

- frequency response

$$H(j\omega) = \frac{\dfrac{j\omega}{RC}}{j\omega^2 + \dfrac{j\omega}{RC} + \dfrac{1}{LC}}$$

- Bandpass Filters

a series RLC BPF	a parallel RLC BPF
	

- General form of the frequency response for BPF

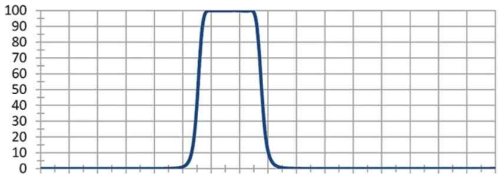
$$H(j\omega) = \frac{constant \times (j\omega)}{(j\omega)^2 + constant \times (j\omega) + constant}$$

$$= \frac{\beta\,(j\omega)}{(j\omega)^2 + \beta\,(j\omega) + \omega_0^2}$$

where $\omega_0 = \sqrt{\dfrac{1}{LC}}$

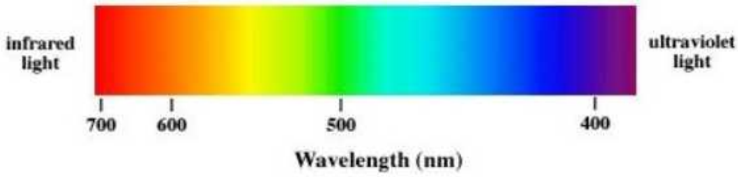
and $\beta = \dfrac{R}{L}$ (series circuit) or $\beta = \dfrac{1}{RC}$ (parallel circuit)

(Example of BPF) Channel selector of TV



(Example of BPF)

choose blue color light from white light

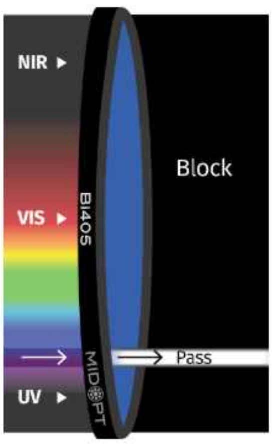


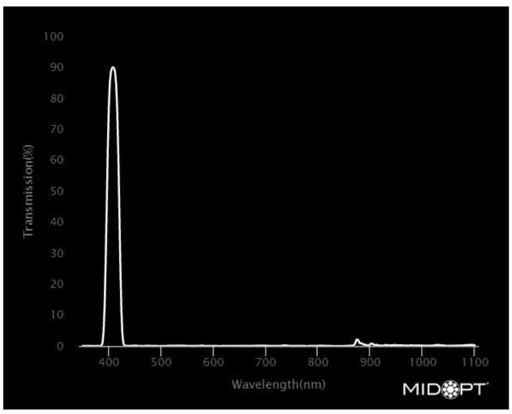
Type of Radiation	Frequency Range (Hz)	Wavelength Range
gamma-rays	10^{20} - 10^{24}	< 10^{-12} m
x-rays	10^{17} - 10^{20}	1 nm - 1 pm
ultraviolet	10^{15} - 10^{17}	400 nm - 1 nm
visible	4 - 7.5*10^{14}	750 nm - 400 nm
near-infrared	1*10^{14} - 4*10^{14}	2.5 μm - 750 nm
infrared	10^{13} - 10^{14}	25 μm - 2.5 μm
microwaves	3*10^{11} - 10^{13}	1 mm - 25 μm
radio waves	< 3*10^{11}	> 1 mm

red color : lower frequency (4×10^{14} Hz = 400 THz)

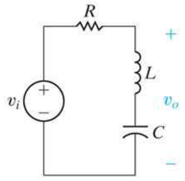
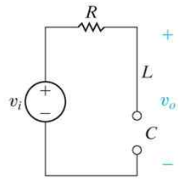
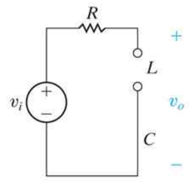
violet color : higher frequency (7.5×10^{14} Hz = 750 THz)

How can we select violet light from the sun light? <u>(Bandpass) Filter</u>

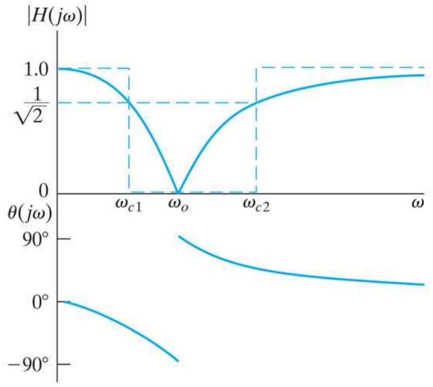




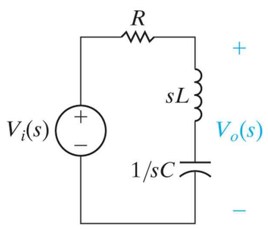
§14.5 Bandreject Filter [BRF] (or Bandstop Filter [BSF])

a series RLC BRF	equivalent circuit ($\omega = 0$)	equivalent circuit ($\omega = \infty$)
		

- Frequency Response of the series RLC BRF



- frequency-domain equivalent of the series RLC BRF



- frequecny response of the series RLC BRF

$$H(j\omega) = \frac{(j\omega)^2 + \dfrac{1}{LC}}{(j\omega)^2 + (R/L)(j\omega) + (1/LC)}$$

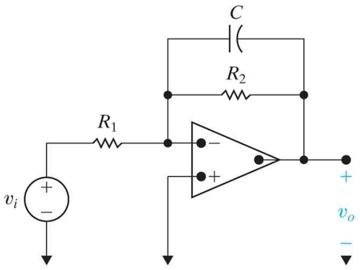
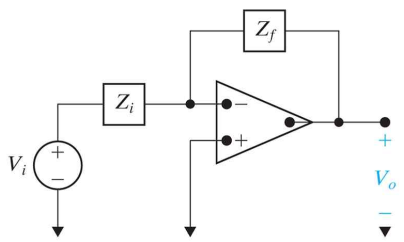
- Summary

LPF	$H(j\omega) = \dfrac{constant}{(j\omega) + constant}$ (#)
HPF	$H(j\omega) = \dfrac{(j\omega)}{(j\omega) + constant}$
BPF	$H(j\omega) = \dfrac{constant \times (j\omega)}{(j\omega)^2 + constant \times (j\omega) + constant}$
BRF	$H(j\omega) = \dfrac{(j\omega)^2 + constant}{(j\omega)^2 + constant \times (j\omega) + constant}$

Chap.15 Active Filter Circuits

※ active : use active elements (op amp)

§15.1 First-Order Active Low-Pass Filter

a first-order active LPF	a general op amp circuit
	

- Frequency Response

$$H(j\omega) = -\frac{Z_f}{Z_i} = -\frac{R_2 \parallel (\frac{1}{j\omega C})}{R_1}$$

$$= -\frac{R_2}{R_1}\frac{\frac{1}{R_2 C}}{j\omega + \frac{1}{R_2 C}} \equiv -K\frac{\omega_c}{j\omega + \omega_c} \qquad (\#\#)$$

K : gain, ω_c : cutoff frequency

※ same type with Eq.(#) or Eq.(##), that is, Low Pass Filter

※ If this frequency response, $\frac{constant}{j\omega + constant}$, is given,

we can have several methods to install in hardware.

- A note about Frequency Response Plots

the transfer function magnitude versus frequency

the transfer function phase angle versus frequency

- Bode plots
  1. Instead of using a linear axis for the frequency value, a Bode plot uses a logarithmic axis.
  2. Instead of plotting the absolute magnitude of the transfer function versus frequency, the Bode magnitude is plotted in decibels(dB) versus the log of the frequency.

- dB (decibel)

  $A_{dB} = 20\log_{10}|H(j\omega)|$ (magnitude ratio)

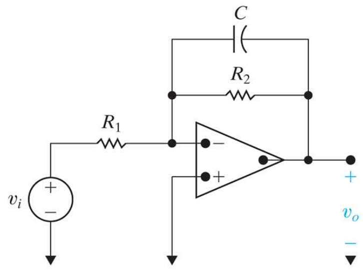
  ※ power ratio : $P_{dB} = 10\log_{10}\dfrac{P}{P_{ref}}$

  amplitude ratio : $A_{dB} = 20\log_{10}\dfrac{A}{A_{ref}}$ (Generally we use this.)

  ※ $20\log_{10}\left|\dfrac{1}{\sqrt{2}}\right| = -3\,dB$ (important)

(example)

Design an active LPF using the circuit ($R_1 = 1\Omega$).



the cutoff frequency : 1 rad/sec

the gain in the passband : 1

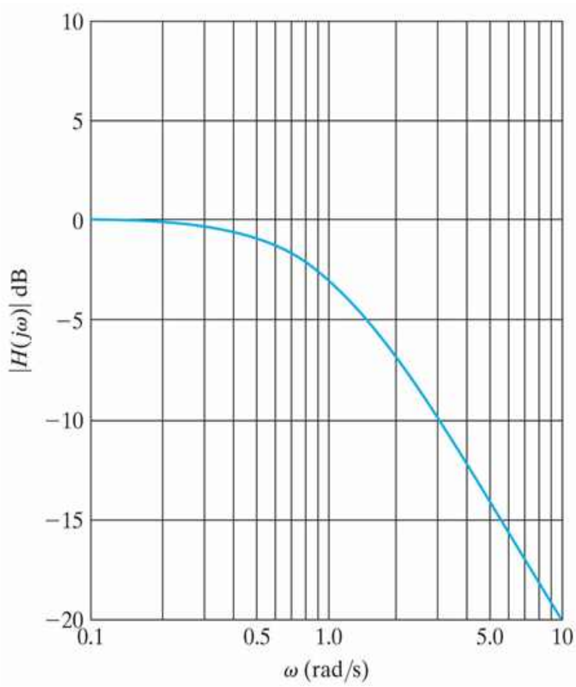
(solution)

$$R_2 = 1\Omega,\ \ C = 1F$$

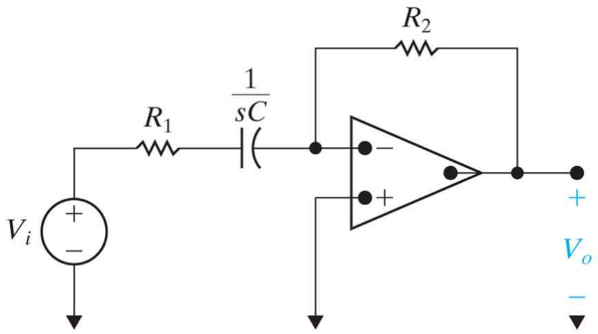
the frequency response

$$H(j\omega) = -\frac{1}{j\omega + 1}$$

the Bode magnitude plot (prototype)



● First-Order Active High-Pass Filter



- Frequency Response

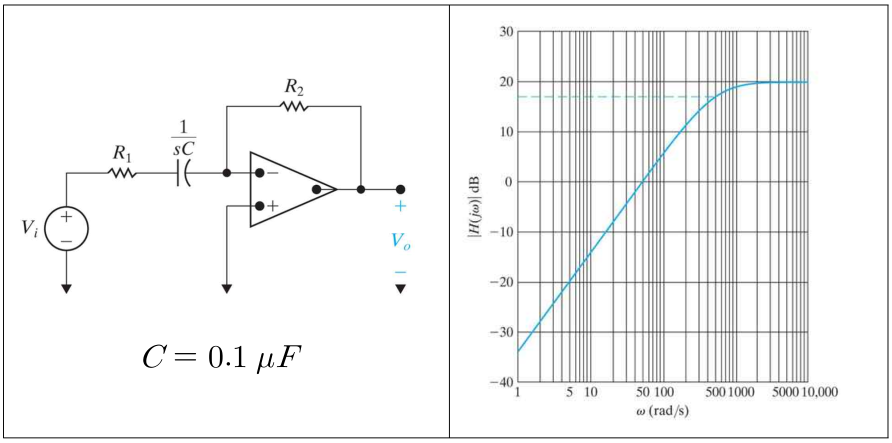
$$H(j\omega) = -\frac{R_2}{R_1 + \frac{1}{j\omega C}} = -K\frac{j\omega}{j\omega + \omega_c}$$

where

$$\text{Gain } K = \frac{R_2}{R_1}$$

and Cutoff frequency $\omega_c = \dfrac{1}{R_1 C}$

(Example) Design an Active HPF



(1) cutoff frequency (-3dB frequency) :

from the graph $\omega_c = 500\ rad/\sec$

$$\omega_c = \frac{1}{R_1 C},$$

so $R_1 = \dfrac{1}{\omega_c C} = \dfrac{1}{500 \times (0.1 \times 10^{-6})} = 20{,}000 = 20k\Omega$

(2) gain in the passband : $A_{dB} = 20$ dB

$$A_{dB} = 20 \log_{10} \frac{V_0}{V_i} = 20 \log_{10} K$$

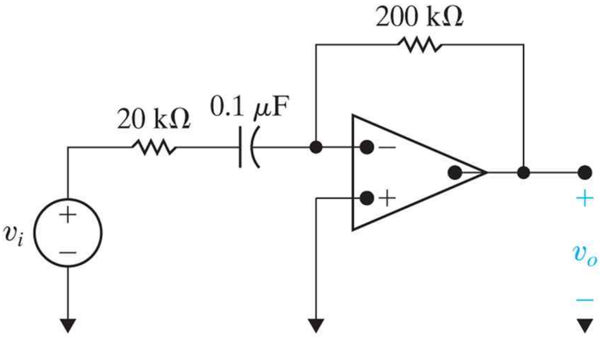
$$\therefore\ K = 10^{\frac{A_{dB}}{20}} = 10^{\frac{20}{20}} = 10^1 = 10$$

$K = \dfrac{R_2}{R_1}$ $\quad \therefore$ Since $R_1 = 20\,k\Omega$, $\quad R_2 = 200\,k\Omega$

Therefore,

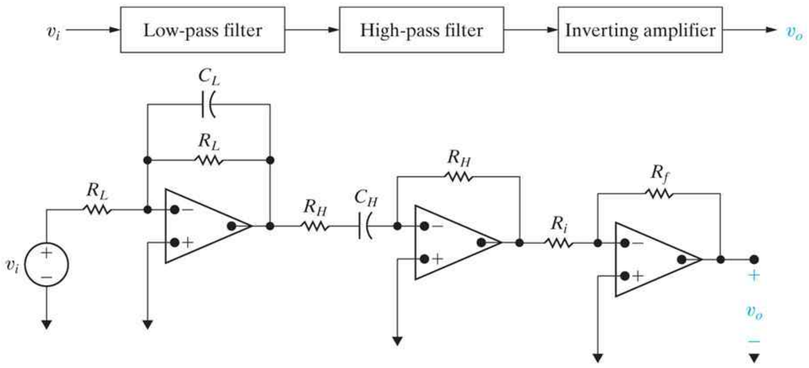
$C = 0.1\ \mu F$, $R_1 = 20\,k\Omega$, and $\quad R_2 = 200\,k\Omega$

The circuit of the active HPF

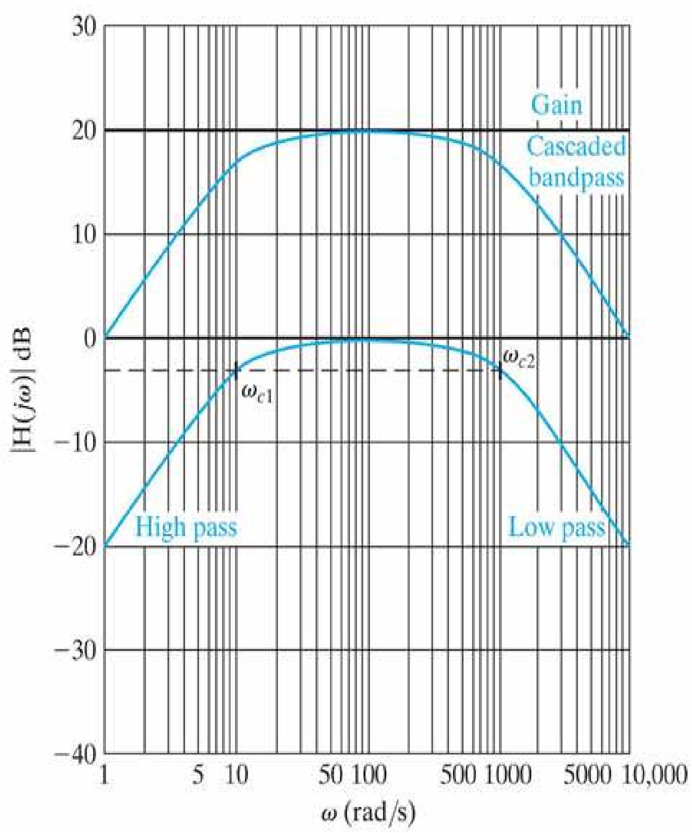


§15.3 Op Amp Bandpass and Bandreject Filters

- Bandpass Filter



- the Bode magnitude plot



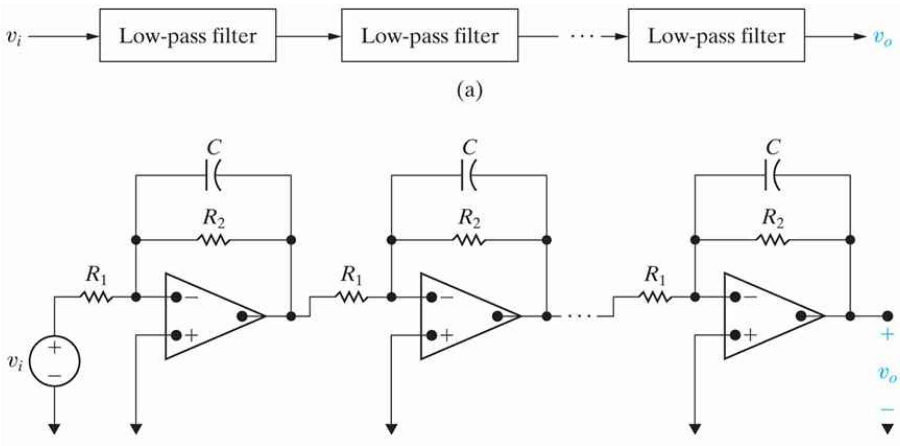
※ broadband

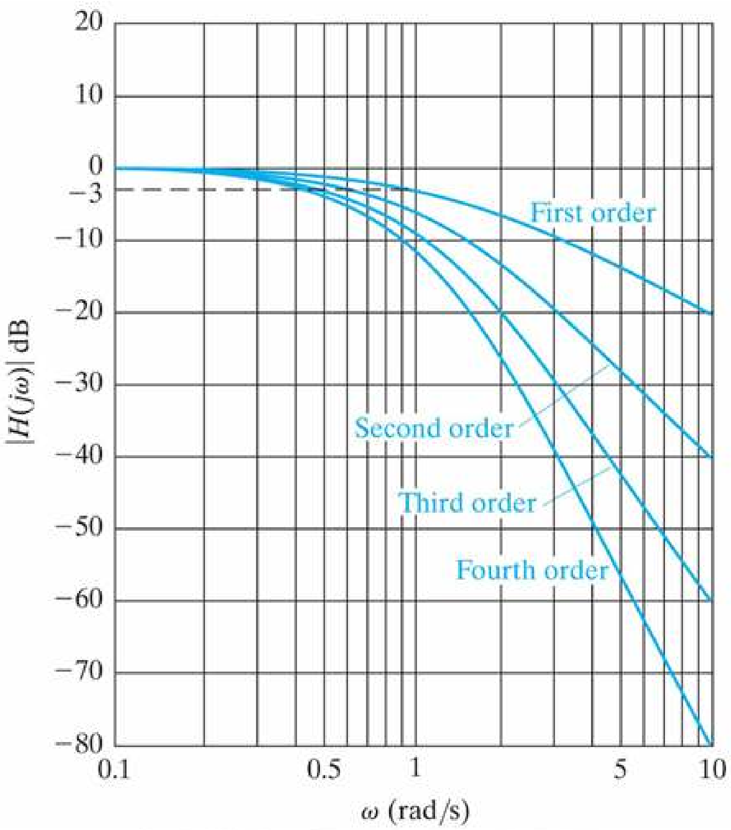
§15.4 Higher Order Op Amp Filters

- Cascading Identical Filters

$$H(j\omega) = \left(\frac{-1}{j\omega + 1}\right)\left(\frac{-1}{j\omega + 1}\right) \cdots \left(\frac{-1}{j\omega + 1}\right)$$

$$= \frac{(-1)^n}{(j\omega + 1)^n}$$

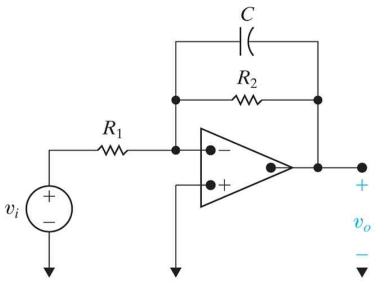
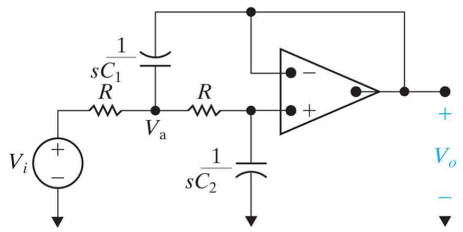




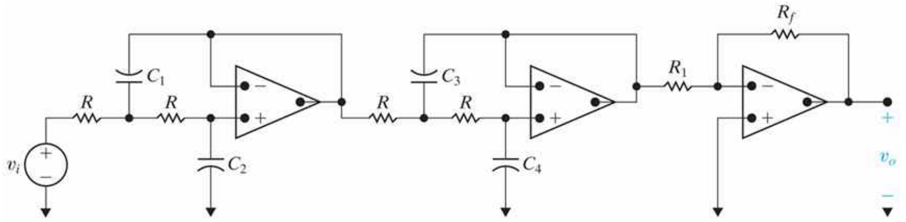
● Butterworth Filter

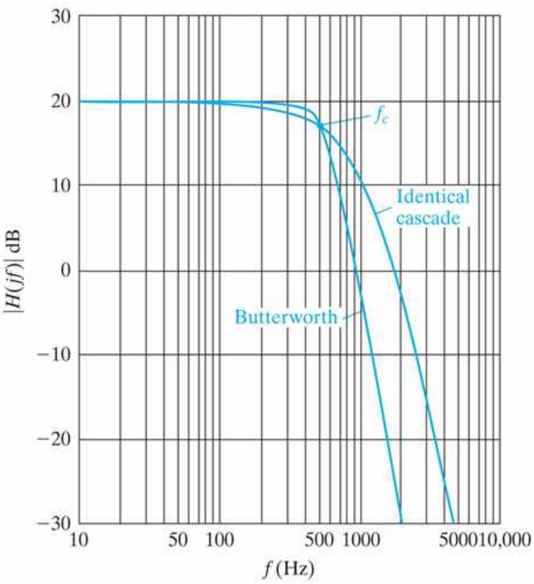
The magnitude of the transfer function of a unity-gain Butterworth LPF

$$|H(j\omega)| = \frac{1}{\sqrt{1+(\frac{\omega}{\omega_c})^{2n}}}$$

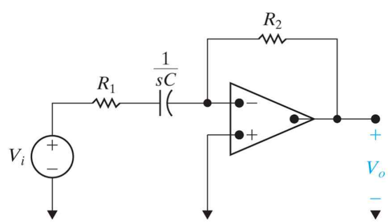
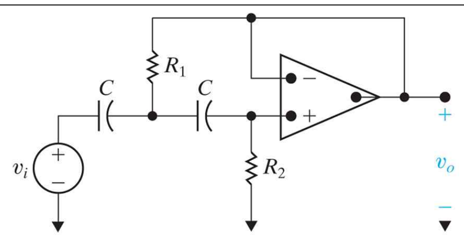
first-order LPF	second-order LPF
	
$H(j\omega) = -\frac{R_2}{R_1}\frac{\frac{1}{R_2C}}{j\omega+\frac{1}{R_2C}}$	$H(j\omega) = \frac{\frac{1}{R^2C_1C_2}}{(j\omega)^2+\frac{2}{RC_1}(j\omega)+\frac{1}{R^2C_1C_2}}$

(example) a 4th-order Butterworth LPF





- Butterworth HPF

first-order HPF	second-order HPF
	
$H(j\omega) = -\dfrac{R_2}{R_1}\dfrac{(j\omega)}{(j\omega)+\dfrac{1}{R_1 C}}$	$H(j\omega) = \dfrac{(j\omega)^2}{(j\omega)^2 + \dfrac{2}{R_2 C}(j\omega) + \dfrac{1}{R_1 R_2 C^2}}$

※ We can connect first- and second-order LPFs and/or HPFs in cascade to make a higher order Butterworth filters.